

Lab 4 (Vibe Coding): Work Summary

Memory Color Pattern game (Pygame), repo SE_PES1UG24AM358, folder Lab-4. **This is a summary written alongside the work. The complete, unedited chat history is submitted separately.**

Task 1: Sequence duplication bug

- Bug: `self.sequence += self.sequence + [new_color]` doubled the sequence every round.
- Fix: `self.sequence.append(new_color)`.
- Second bug found during testing: the 150 ms click-flash timer from the previous round's last click switched off the first pad of the new playback. Fixed by clearing `player_lit_button` at the start of `start_next_round()`.
- Verified: rounds 1, 2, 3 show exactly 1, 2, 3 flashes.

Task 2: Playback acceleration

- `flash_duration = max(180, 450 - score*40)`; `pause_duration = max(80, 200 - score*20)`.
- Computed from score inside `start_next_round()`, so `reset()` (score = 0) restores 450/200 ms.
- Reached the 180/80 ms floors by round 8 in a simulated run.

Task 3: Sound tones

- Sine tones synthesized in memory (no files): Red 261 Hz, Blue 329 Hz, Green 392 Hz, Yellow 523 Hz.
- 20 ms fade in/out to avoid clicks; all tones stopped before a new one plays so they never overlap.
- Plays on playback flashes and on player clicks; falls back to silence if the mixer cannot start.

Task 4: Countdown timer

- 3000 ms allowed per click; bar below the pads shrinks and changes green, yellow, red.
- Timer restarts at the start of the player turn and after each correct click.
- Timeout ends the game with a 'TIME'S UP' message; R resets everything.

Problems hit along the way

- Mixed tabs and spaces from Notepad edits caused `TabError`; fixed by using generated tab-free files.
- Commands run from the wrong folder or with a wrong destination path; corrected.
- A leftover debug print statement had been left in the file; removed in the clean versions.

Critical review points

- Timeout uses `pygame.time.get_ticks()` at 60 FPS, so accuracy is about one frame (roughly 16 ms).
- Stopping tones before each new one avoids overlap at the fastest 180 ms flash speed.
- The timer is reset in `register_player_click` so each click gets a fresh limit.
- Tested with a headless simulation (rounds 1 to 8, reset, timeout, timer reset) and then manually by playing.